

Defining Family-Centered AI Through the Circulation of Care

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ABSTRACT

Existing accounts of family-centered AI often position middle-aged adults as primary mediators of AI-related care in two generational roles. As parents, they monitor, interpret, and guide their children's AI use as a form of protective and educational care. As adult children of older parents, they use AI to coordinate and provide care for aging relatives. Although this "sandwich generation" framing captures important forms of labor, it can obscure how care, information, and technological mediation circulate across family relationships rather than consistently flowing from a single generation. Grandparents may provide emotional, practical, and technological support to their adult children and grandchildren, while younger family members may act as information or technology brokers for other family members. Family members may therefore move between providing and receiving care as needs and capacities change. I consequently define family-centered AI as AI embedded in the circulation of care across interdependent family relationships, where one member's use can shape the support, responsibilities, decisions, and experiences of others.

1 POSITION STATEMENT

Emerging research on AI in families has largely examined care across two age-defined relationships. In research concerning children, parents are commonly positioned as responsible for monitoring, interpreting, and guiding children's AI use. Children encounter AI across a growing range of everyday contexts, including smart home devices [1], chatbot [12], and digital games [7]. From my past research with children, I also observed children critically interpreting AI-mediated game content; for example, one child avoided games with apparently AI-generated covers because they expected them to be "AI slop." Although children may develop their own judgments about such content, their encounters with AI can also raise concerns about quality, misinformation, manipulation, and safety that draw parents into monitoring and guiding their engagement [12]. This involvement can be understood as a form of protective and educational care. For example, Xie et al. studied parents aged 31–45 and children aged 7–13, describing parents as supervising children's interactions with AI, reinterpreting its outputs, and gradually transferring responsibility as children developed greater independence [10].

At the other end of the life course, research on AI-supported family caregiving commonly positions middle-aged adults as responsible for older relatives. Yee et al., for instance, recruited caregivers aged 45–64 who provided care to adults aged 65 or older; participants had a mean age of 56.7, and most cared for a parent [11]. A recent AARP account illustrates how these age categories can blur in practice: when her 94-year-old mother began showing signs of vascular dementia, 61-year-old adult child used AI to interpret medical terminology, research home-safety technologies, and create visual schedules to support her mother's daily life [9]. Although the adult child could be categorized as either middle-aged or older

depending on the age thresholds adopted, she remained an active provider of AI-supported care. This case suggests that age alone does not determine whether a family member provides or receives care. Organizing family-centered AI around fixed life-stage roles can therefore reproduce a model in which a middle generation provides care downward to children and upward to older relatives, overlooking how caregiving roles may persist, overlap, and shift across the life course.

Care within families does not follow a fixed caregiver–recipient hierarchy but circulates as members’ needs, capacities, and circumstances change. Older parents may continue to provide adult children with emotional support, practical assistance, and advice [2], while grandparents use digital communication technologies to sustain relationships and participate in care across geographic distance [3, 6]. Younger family members may likewise act as information and technology brokers, helping parents and grandparents access digital services, communicate online, and interpret unfamiliar systems [4]. These exchanges may be reciprocal without being equal or simultaneous: a family member may provide technological assistance in one context while receiving emotional, practical, or informational support in another. Care is therefore relational and situational rather than determined solely by age or generation. Recognizing this circulation provides a foundation for examining how AI becomes embedded in, and potentially reorganizes, existing family care practices.

However, the circulation of care does not mean that family members have equal capacities to use AI critically when supporting others. Empirical research identifies challenges across age groups. Children may initially overtrust generative AI and revise their judgments only after encountering its errors [5], while parents responsible for mediating teenagers’ AI use may struggle to understand their children’s practices and the associated risks [12]. Research with adults over 60 has similarly documented difficulties identifying misinformation during AI use [8]. These findings raise the possibility that willingness to provide care does not necessarily entail the capacity to critically evaluate AI-generated information. When one family member uses AI to support another or supports that person’s use of AI, such limitations may introduce inaccurate or poorly contextualized claims into family interactions and shift verification and interpretive labor onto other family members.

Building on these observations, I argue that family-centered AI should be understood as AI embedded in the circulation of care across interdependent family relationships. At the workshop, I hope to discuss how this perspective can broaden definitions of family-centered AI by considering who provides and receives AI-related care, who assumes responsibility for interpreting or responding to AI outputs, and how family members who are not directly involved in an interaction may nevertheless be affected by it.

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